$$p_x(t) = \int \mathrm{d}\epsilon\, J(\epsilon) \frac{\Delta_0^2 + \epsilon^2 \cos\left(2\Omega_\epsilon t\right)}{\Omega_\epsilon^2} p_x(0), \tag{5.166a}$$

$$p_y(t) = \int \mathrm{d}\epsilon\, J(\epsilon) \left[\cos\left(2\Omega_\epsilon t\right) p_y(0) - \frac{\Delta_0 \sin\left(2\Omega_\epsilon t\right)}{\Omega_\epsilon} p_z(0)\right], \tag{5.166b}$$

$$p_z(t) = \int \mathrm{d}\epsilon\, J(\epsilon) \left[\frac{\epsilon^2 + \Delta_0^2 \cos\left(2\Omega_\epsilon t\right)}{\Omega_\epsilon^2} p_z(0) + \frac{\Delta_0 \sin\left(2\Omega_\epsilon t\right)}{\Omega_\epsilon} p_y(0)\right], \tag{5.166c}$$